```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv("/cancer patient data sets.csv")

# Quick look
print(df.shape)
df.head()
```

```
(1000, 26)
```

	index	Patient Id	Age	Gender	Air Pollution	Alcohol use	Dust Allergy	OccuPational Hazards	Genetic Risk	chronic Lung Disease	...	F
0	0	P1	33	1	2	4	5	4	3	2	...	
1	1	P10	17	1	3	1	5	3	4	2	...	
2	2	P100	35	1	4	5	6	5	5	4	...	
3	3	P1000	37	1	7	7	7	7	6	7	...	
4	4	P101	46	1	6	8	7	7	7	6	...	

5 rows × 26 columns

```
# Info & summary
df.info()
df.describe()

# Check missing values
df.isnull().sum()

# Unique values in categorical columns
df['Level'].value_counts()
df['Gender'].value_counts()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 26 columns):
 #   Column                    Non-Null Count  Dtype 
---  ------                    --------------  ----- 
 0   index                     1000 non-null   int64 
 1   Patient Id                1000 non-null   object
 2   Age                       1000 non-null   int64 
 3   Gender                    1000 non-null   int64 
 4   Air Pollution             1000 non-null   int64 
 5   Alcohol use               1000 non-null   int64 
 6   Dust Allergy              1000 non-null   int64 
 7   OccuPational Hazards      1000 non-null   int64 
 8   Genetic Risk              1000 non-null   int64 
 9   chronic Lung Disease      1000 non-null   int64 
 10  Balanced Diet             1000 non-null   int64 
 11  Obesity                   1000 non-null   int64 
 12  Smoking                   1000 non-null   int64 
 13  Passive Smoker            1000 non-null   int64 
 14  Chest Pain                1000 non-null   int64 
 15  Coughing of Blood         1000 non-null   int64 
 16  Fatigue                   1000 non-null   int64 
 17  Weight Loss               1000 non-null   int64 
 18  Shortness of Breath       1000 non-null   int64 
 19  Wheezing                  1000 non-null   int64 
 20  Swallowing Difficulty     1000 non-null   int64 
 21  Clubbing of Finger Nails  1000 non-null   int64 
 22  Frequent Cold             1000 non-null   int64 
 23  Dry Cough                 1000 non-null   int64 
 24  Snoring                   1000 non-null   int64 
 25  Level                     1000 non-null   object
dtypes: int64(24), object(2)
memory usage: 203.3+ KB
```

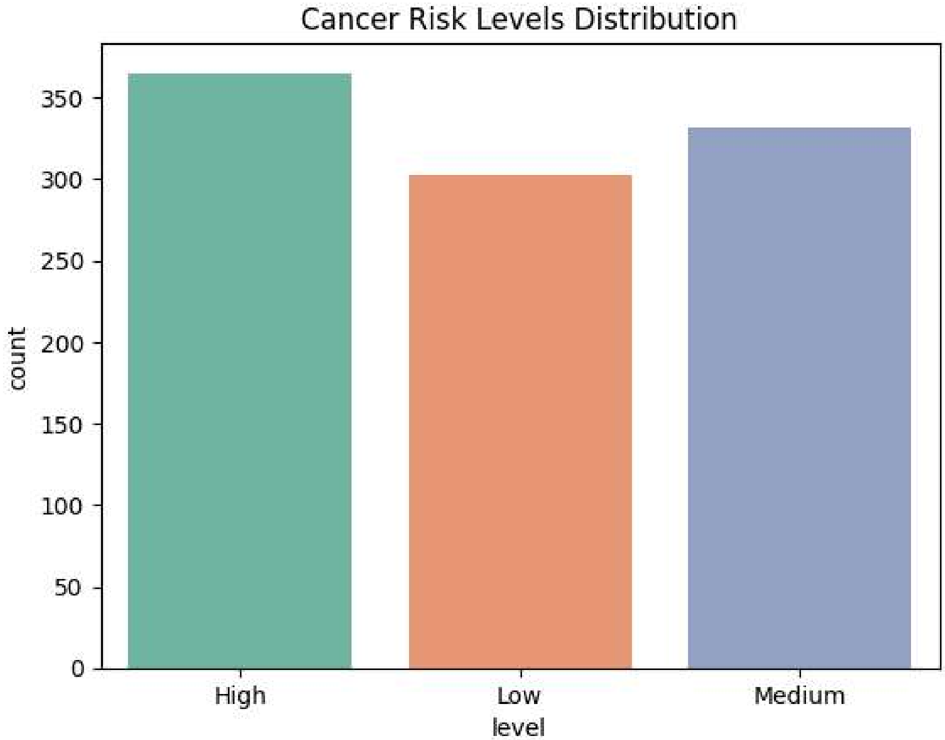
	count
Gender	
1	598
2	402

dtype: int64

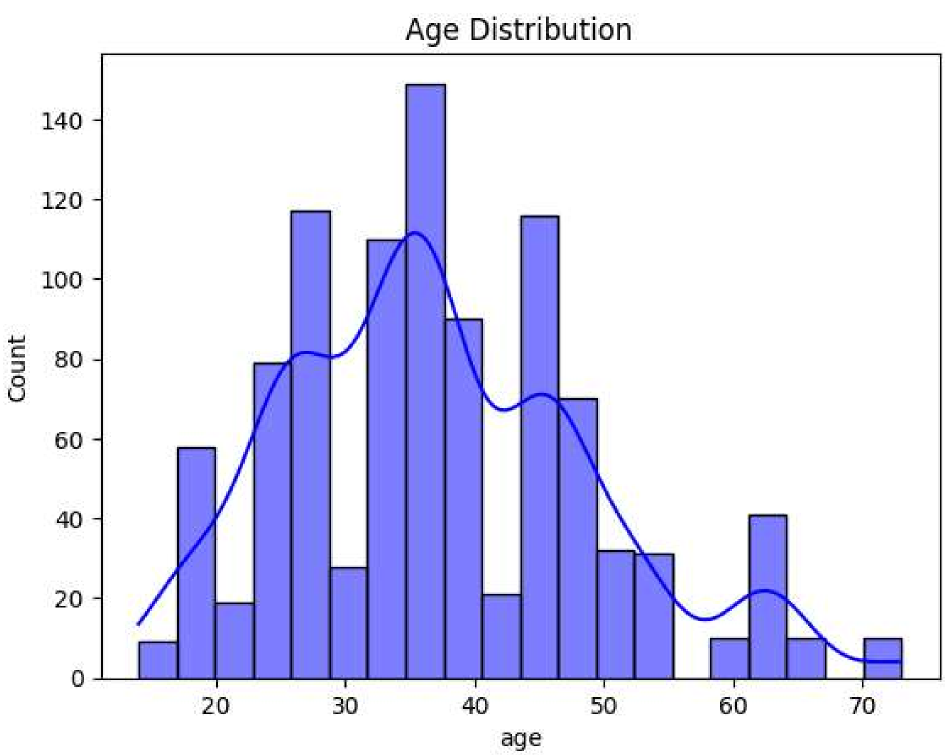
```
df.columns = df.columns.str.strip().str.lower().str.replace(" ", "_")

df['gender'] = df['gender'].map({1:'Male', 2:'Female'})
df['level'] = df['level'].astype('category')
```

```
sns.countplot(x='level', data=df, palette='Set2')
plt.title("Cancer Risk Levels Distribution")
plt.show()
```

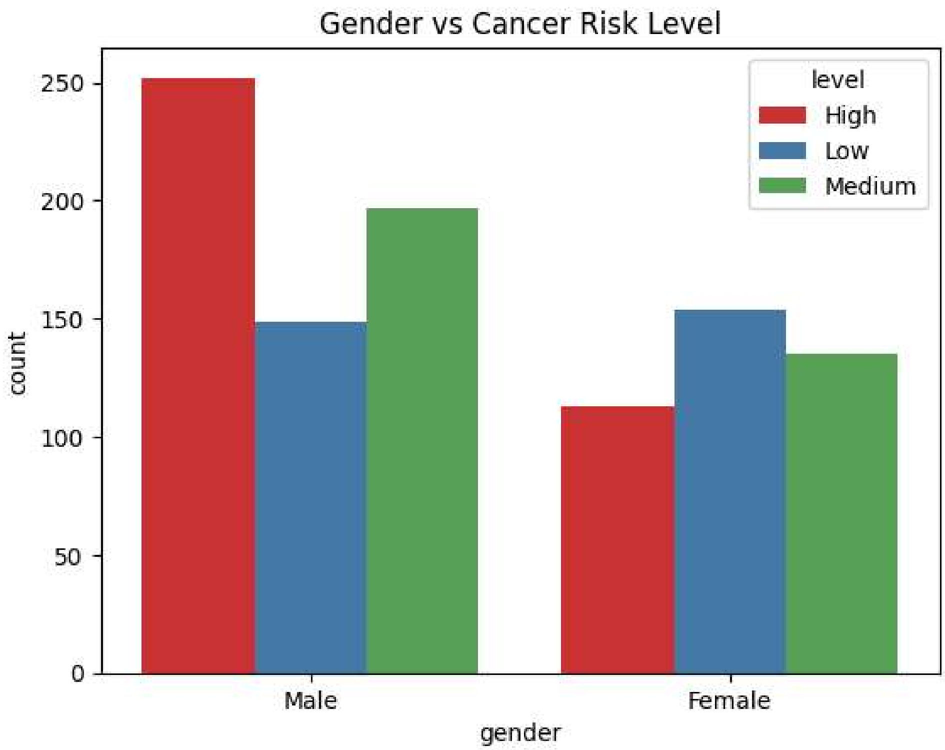
```
/tmp/ipykernel_777/3814147754.py:1: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the

  sns.countplot(x='level', data=df, palette='Set2')
```




```
sns.histplot(df['age'], bins=20, kde=True, color='blue')
plt.title("Age Distribution")
plt.show()
```

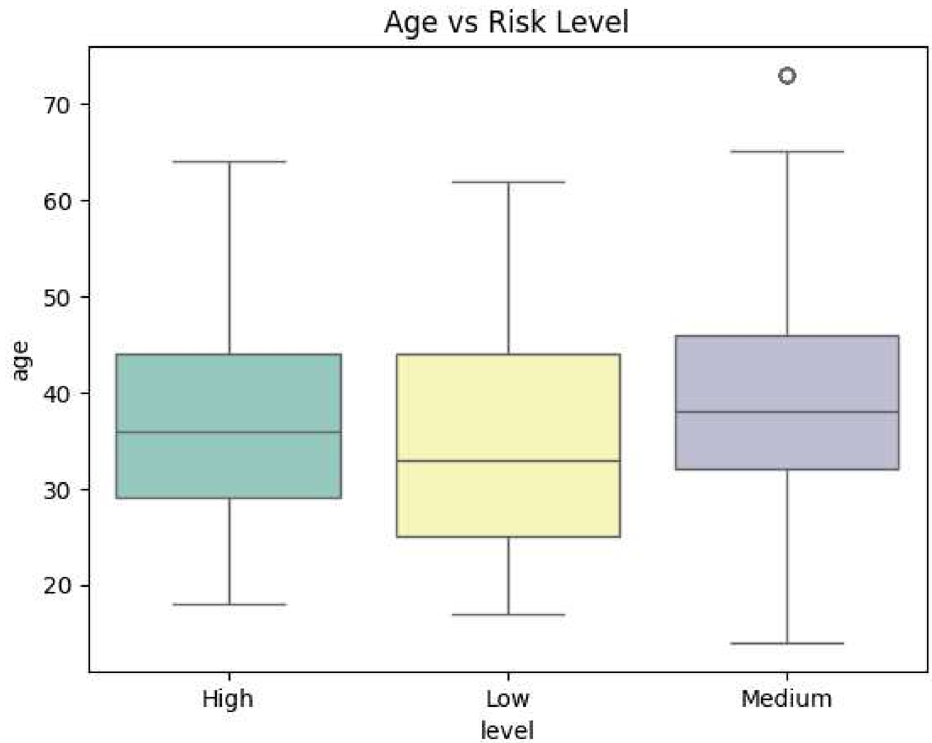


```
sns.countplot(x='gender', hue='level', data=df, palette='Set1')
plt.title("Gender vs Cancer Risk Level")
plt.show()
```

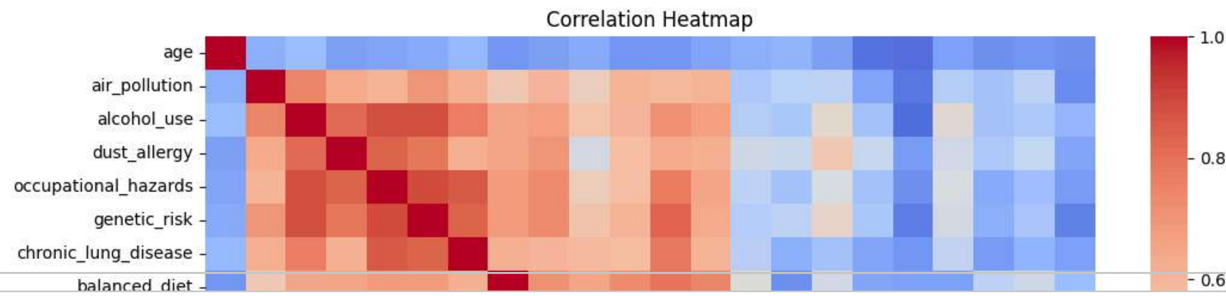



```
sns.boxplot(x='level', y='age', data=df, palette='Set3')
plt.title("Age vs Risk Level")
plt.show()
```

```
/tmp/ipykernel_777/552579712.py:1: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the

  sns.boxplot(x='level', y='age', data=df, palette='Set3')
```




```
plt.figure(figsize=(12,8))
sns.heatmap(df.drop(columns=['patient_id','index']).corr(numeric_only=True), cmap='coolwarm', anr
plt.title("Correlation Heatmap")
plt.show()
```




```
group_means = df.groupby('level').mean(numeric_only=True)
plt.figure(figsize=(12,6))
```